Data cleaning and preparation

Missing and empty data

Before class begins…

- Read Chapter 8 in your text.

Missing and empty values

We talked earlier about the difference between missing and empty values. What do we do about them?

Some algorithms can ignore them (Naïve Bayes, for one), and some can't. If there are many missing values in an attribute, you are probably better off deleting it entirely. But what if you need to keep it?

We've got some questions about missing values.

1. What do they mean?
2. Where do they come from?
3. How do they impact the mining project?
4. What do we do about them?

Missing/empty types

Missing value is..	Example	Source
Unknown	Respondents don’t answer survey question; in automated data collection systems, data dropout	Survey design/business goal; for automated systems, sensor failure
Null	Activation date for a credit card, before the card is activated	Database design
Error	Feeding nominal data into a numeric field	Transforming or transferring data between platforms or systems
Undefined	Dividing by zero; mathematical operations on other missing values	Automated data analysis systems

Missing value information

Sometimes the pattern of missing values is important to us, because it can indicate problems with our data collection, data entry, file merge, or required fields that we need to address. If you are able to influence collection methods, doing a pilot study will point these problems out.

Beware of data set descriptions that say “No missing values” but populate the dataset with values that make no sense in that context (for instance, the zero value for diastolic blood pressure in the diabetes dataset).

Basic strategies: First, do no harm

The main thing to remember is not to bias the dataset any more than is necessary.   Here are some strategies to deal with these issues:

1. **Substitute a value.**
2. **Predict the value.**
3. **Impute the value.**
4. **Remove the record.**
5. **Discard the variable.**

Note that the first three methods attempt to fill in the holes in the data; the last two are row and column deletions.

Basic strategy 1: Substitute a value

Here we fill in the missing spot with a value of our choosing, either one based on the characteristics of the data, or a user-defined value to denote missing entries. What are our options?

- **Nominal data**
  - The mode (most occurring value)
  - Any of the categories (user-defined)
  - A new category that indicates a missing value
- **Ordinal and continuous data**
  - The mean (for continuous data), median, or mode
  - Standard deviation-preserving value
  - Maximum or minimum value
  - Zero or other user-defined value

Substituting values by variable (column-wise replacement)

A careful examination of each variable's descriptive statistics will give you some idea of what would be the least disruptive value to put in.
If you are concerned about maintaining the variable's central tendency, use the mean. HOWEVER, if there's a big difference between the mean and the median, you should probably use the median. Remember that a large difference between those two measures signals the presence of outliers.

Other strategies include controlling for variation by using a value that preserves the standard deviation of the variable. Pyle's experiments show that method seems to do the least harm to the dataset. You can also use a global constant, but that usually introduces bias.

Substituting values by class relationship (row-wise replacement)

An additional strategy is to examine the class value for the particular record with the missing value, and choose a replacement based on the class statistics.

For example, if the target class value is “play” (assuming the weather dataset), then we would replace a missing temperature value with the mean (or median, or standard deviation-preserving value) of the class “play.”

This approach supports the relationship of the variable with the class attribute.

Strategy 2: Predict the value

If I know miles per gallon (mpg) varies inversely with weight, I can predict a likely value for mpg given the weight of a car, and vice versa. This approach uses linear regression (for one, or multiple values). The idea is to fit the data to a line corresponding to the equation

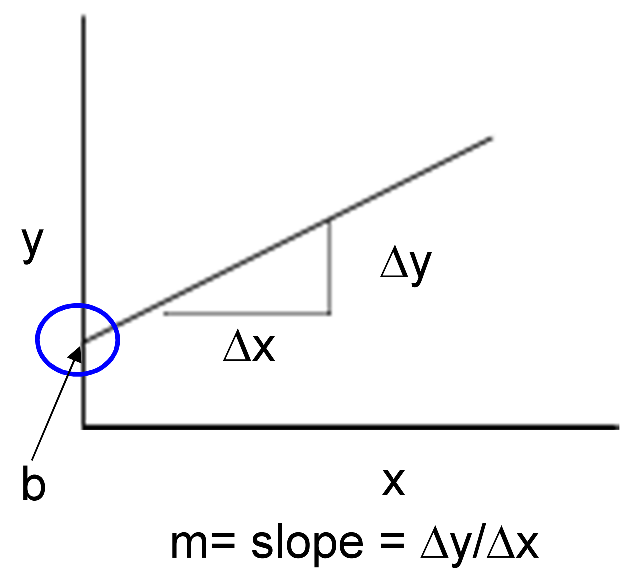
$$Y=mx+b$$

We call *y* the dependent variable and *x* the independent variable, because the value of *y* depends on the value of *x*.

More on linear regression

Linear, or least squares, regression attempts to fit a line to the data that describes the relationship between the two variables. The line shows how *y* changes with each unit change of *x*.

Recall from algebra that the equation of a straight line is *y=mx+b*, where *m* is the slope and *b* is the point where the line intersects the *y* axis. The slope is calculated as the change in y divided by the change in x. Sometimes you hear this fraction, or ratio, called “rise over run”.



Equation of the least-squares line

The least-squares line equation is very similar to the algebraic one. S_y is the standard deviation of the y values, while S_x is the standard deviation of the x values.

Equation of the Least-squares Regression Line

The equation of the least-squares regression line is given by $\hat{y} = b_1 x + b_0$

$b_1 = r \cdot \dfrac{s_y}{s_x}$ is the **slope** of the least-squares regression line

$b_0 = \overline{y} - b_1 \overline{x}$ is the **intercept** of the least-squares regression line

Recall that

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

Fitting a line in Minitab

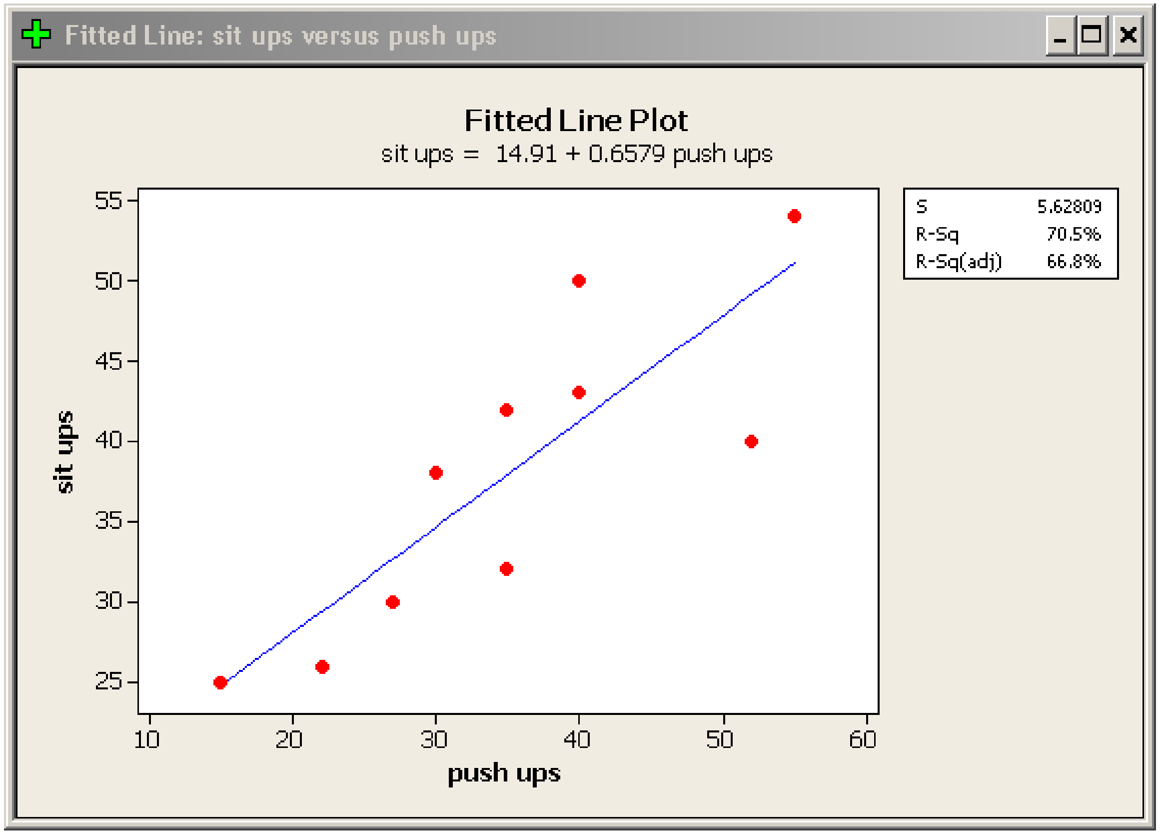
To fit a regression line in Minitab, choose

Stat > Regression > Fitted Line Plot

Suppose we consider this dataset, where students performed as many push-ups and sit-ups as they can in one minute. We want to predict how many sit-ups a student can do, based on the number of push-ups.

Student	Push-ups	Sit-ups
1	27	30
2	22	26
3	15	25
4	35	42
5	30	38
6	52	40
7	35	32
8	55	54
9	40	50
10	40	43

Line plot from Minitab



Some points to consider

- The slope b_1 (+0.66) represents the predicted change in *y* per unit increase in *x*. This concept is important to understand. In our example, for each push-up a person can do, how many more sit-ups will he be able to do? For every additional push-up, a person can do 0.66 extra sit-ups. What would be the relationship between sit-ups and push-ups if we had a negative slope?
- In this problem the *y* intercept–where the line crosses the *y*-axis (and where *x*=0) is 14.9. Is this number meaningful in this analysis? If a person cannot do any pushups (x=0) would he be able to do almost 15 sit-ups? Probably not. So, for this case the *y* intercept is meaningless.

More points

- Using the equation (also called the model) to predict *y* within the range of the data is called "interpolation" and is, statistically, fine to do. Prediction outside the data set (extrapolation) is risky and may not be accurate. For this dataset, *X* ranges from 15 to 55 and *Y* ranges from 25 to 54. Consider the *y*-intercept again; *x*=0 is outside the dataset range, so trying to predict the number of situps when *x*=0 would be extrapolating—and clearly, in this case, not accurate.
- The regression equation will always pass through the centroid (x-bar,y-bar), which is the average of all x values and the average of all y values.

Multiple linear regression

In multiple linear regression, we have many independent variables that affect the value of the dependent variable. The equation is that of a plane or hyperplane, where α is the intercept, and β_i is a coefficient.

$$\hat{y} = \alpha + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

The relative size of the coefficients tell you how important that particular variable is in predicting *y.* Another method that is very similar to MLR is analysis of variance (ANOVA). Both methods assume a linear relationship, as does linear regression.

Strategy 3: Impute the values

If we must, we can fill in the missing data by guessing the values. This process is called *imputation* and should always be used with caution. The data values generated by these methods are **estimates** and we should never lose sight of that.

The most naïve method is to simulate a distribution from the non-missing values, and use that distribution as a source. For example, suppose we had the following data:

1,3,4,2,6,6,3,,,1

We have eight values and three missing values. Given the existing data, the probability of 1 occurring in this list is [P(1) = 2/8]; the probability of 2 occurring is [P(2) = 1/8], P(3) = 2/8, etc.

Imputing data

We can assume the missing values follow the same distribution, and impute those values using these probabilities. While this approach is very naïve indeed, and makes probably incorrect assumptions, it's easy to implement, cheap to run, and simple to interpret.

Multiple imputation

The naïve method imputes only one value for each missing entry. However, a more sophisticated approach imputes multiple values for each entry, generating multiple data sets (one for each imputed value). The datasets are individually analyzed, and the results combined.

This method assumes the attributes are missing monotonically—which means that the attributes can be ordered so that if attribute *j* has missing values, all attributes 1 through *j*-1 are NOT missing.

You can see that best using an example.

Monotonically missing

The initial dataset has five variables and ten records. We can permute the columns to attempt to make this dataset's missingness pattern monotonic.

Record	X1	X2	X3	X4	X5
1	-	-	-	-	-
2	13	10	18	12	13
3	18	16	12	12	11
4	-	-	-	12	4
5	18	7	5	16	11
6	1	17	6	19	4
7	-	8	-	-	7
8	-	8	-	13	11
9	14	-	-	16	18
10	-	-	-	9	7

Now the dataset has a monotone missing data pattern, and we can use multiple imputation techniques to fill in the values.

Record	X5	X4	X2	X1	X3
1	-	-	-	-	-
2	13	12	10	13	18
3	11	12	16	18	12
4	4	12	-	-	-
5	11	16	7	18	5
6	4	19	17	1	6
7	7	-	8	-	-
8	11	13	8	-	-
9	18	16	-	14	-
10	7	9	-	-	-

A small example

Suppose we have a monotone missing data pattern like this:

X1	X2	X3	X4
20	2	10	5
15	5	9	3
25	5	10	-
20	4	-	-
10	1	-	-

We recursively use regression to impute the data values, going from left to right. So the first model would be

$$X3 = a + b_1 X1 + b_2 X2$$

The result (plus a random error term) replaces the missing values, and we then generate values for X4 using this model:

$$X4 = a + b_1 X1 + b_2 X2 + b_3 X3$$

When we are done, we will have multiple datasets; we combine those to generate a single dataset.

Choices for imputation methods

There has been a lot of research on imputation methods, so I'm not going to go into **all** the different algorithms available. But here is a table that gives you some idea of what type of algorithm to use when.*

Pattern of missingness	Type of imputed variable	Recommended method
Monotone	Continuous	Regression Predicted mean matching Propensity score
Monotone	Ordinal	Logistic regression
Monotone	Nominal	Discriminant function method
Arbitrary	Continuous	MCMC full-data imputation MCMC monotone-data imputation

*Refaat, p. 182

Monte Carlo Markov Chain (MCMC)

When the pattern of missingness is not monotone, we use MCMC methods to impute the missing values. However, MCMC methods are expensive and infeasible for large datasets, so we may only want to use them to impute enough values to make the pattern monotone, then switch to the easier regression methods.

In MCMC, the data are assumed to be from a multivariate normal distribution. The method has three steps:

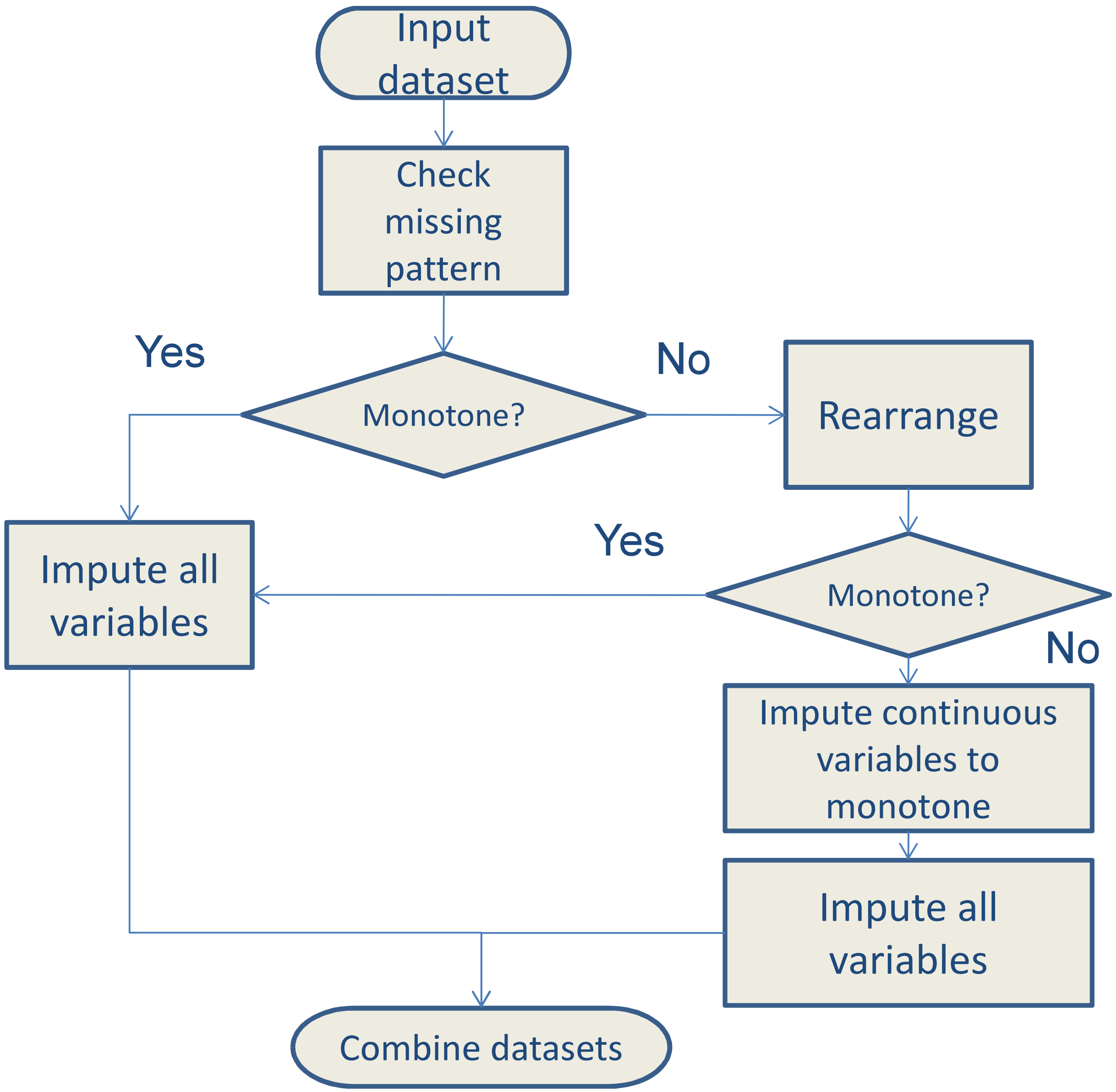
1. Build the conditional distribution of the missing values from the observed values to estimate the data
2. Compute the parameters of a multivariate normal distribution using the filled-in dataset
3. Repeat 1 and 2 until the estimates are stable.

Why it's Markov chain

Each iteration of the MCMC algorithm results in a set of estimated values and a set of parameters for the distribution. The values depend only on the $(i\text{-}1)^{th}$ iteration. (The method is called Monte Carlo because we do this many times, with different seeds, to generate multiple values.)

SAS has a robust multiple imputation implementation. R has a package called MICE that performs these functions.

The multiple imputation process



Strategy 4: Remove the record

Here, we remove the record with the missing values. The question you should be asking is how many values should be missing before we remove a record? There's no right answer—it depends very much on which attributes are involved.

Remember, though, that most datasets are very sparse; if we removed every record with a missing value, we might not be left with any records at all. Also, the scoring data will also have missing values. If we train on a full dataset, the scoring data, with its missing entries, may not be correctly classified.

Removing records

If class attribute is missing, then removing the record is an obvious step—we can't use a record for training that doesn't contain the class value. Could we use it for testing a model? Why or why not?

Data resulting from some scientific and engineering domains usually come from automated collection systems. If entries are missing in those systems, then something has gone wrong with the data collection equipment. In those cases, we should probably not use that data.

Strategy 5: Discard the variable

When should we discard a variable? A common rule is any variable with 20% of its values missing. But that would take out a large number of attributes in most data mining sets. So we set the bar higher—approximately 80-90% missing values would indicate the variable isn't going to be useful.

Any time you discard variables or remove records, you remove information (obviously, not if the variable is a constant or completely missing, or the record has no data entries).

Try different replacement methods with this weather data

Outlook	Temperature	Humidity	Wind	Play
Sunny	85	85	FALSE	NO
Sunny	?	90	TRUE	NO
Overcast	83	86	FALSE	YES
Rainy	70	96	FALSE	YES
Rainy	68	80	FALSE	YES
Rainy	65	?	TRUE	NO
Overcast	64	65	TRUE	YES
Sunny	72	95	FALSE	NO
Sunny	69	70	FALSE	YES
Rainy	75	?	FALSE	YES
Sunny	75	70	TRUE	YES
Overcast	?	90	TRUE	YES
Overcast	81	75	FALSE	YES
Rainy	71	91	TRUE	NO

Here is the complete dataset

Outlook	Temperature	Humidity	Wind	Play
Sunny	85	85	FALSE	NO
Sunny	80	90	TRUE	NO
Overcast	83	86	FALSE	YES
Rainy	70	96	FALSE	YES
Rainy	68	80	FALSE	YES
Rainy	65	70	TRUE	NO
Overcast	64	65	TRUE	YES
Sunny	72	95	FALSE	NO
Sunny	69	70	FALSE	YES
Rainy	75	80	FALSE	YES
Sunny	75	70	TRUE	YES
Overcast	72	90	TRUE	YES
Overcast	81	75	FALSE	YES
Rainy	71	91	TRUE	NO

Summary

When you choose to replace missing values, remember that there is no substitute for thinking. Don’t just blindly run a filter—make sure you know exactly what is happening with your data. Also, keep a log of what you’ve done with your data. With Big Data, there’s no way you are going to keep multiple copies of your dataset, but you should at least be able to recreate the changes you’ve made along the way (or undo them!).

The point of manipulating data is to tease out the information that is hidden in the dataset.

Homework 4: Due **4/14**

Try different ways to fill in the missing values in the diabetes dataset. Use a decision tree method in either Weka or Rattle (or the data mining software of your choice*) on the original dataset and then your filled-in dataset. What are the differences? Did you improve the results?

* If you need some help with Weka, let me know.
You must change the class variable from {0,1} to {No, Yes} to use J48.

Discussion for this week

- Discussion question: Find a recent (what do you consider recent?) research article about missing data techniques, and post your review.